

How To Create A Pro Tools Friendly AAF From Adobe Premiere

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In this article, we explain how to export a Pro Tools friendly AAF from Adobe Premiere Pro.

Although the basic process is relatively simple, there are some complications if the video editor has used nested clips and sequences, merged clips or you need to get the metadata from the files into Pro Tools to use with the Field Recorder Workflow.

What Is the AAF format?

AAF stands for the Advanced Authoring Format. It is a technology designed specifically for digital nonlinear post-production and authoring. You can consider it as a "Super-EDL," but actually it is also much more.

The AAF solution has several components: It supports the complex combination of a piece of content and its associated metadata. This allows the content to be described as a media object. It facilitates the cross-platform interchange of metadata or program content.

What is the difference between the OMF and AAF formats?

Open Media Framework (OMF) or Open Media Framework Interchange (OMFI) is a platform-independent file format intended to transfer digital media between different software applications.

AAF (and OMF) are used to transfer sequence/timeline information from one application to another. OMF is supported in a number of applications,

including AVID and Pro Tools.

AAF is the newer format and contains more information than the OMF. For example, OMF drops the volume automation and names of the tracks when exporting and importing from one application to another. AAF retains the volume automation and the track names. Most applications for audio and video support AAF and MXF formats moving forward.

When it comes to OMFs and Premiere Pro, [Adobe](#) states that...

"The OMF files exported by Premiere Pro are not officially supported for platforms other than ProTools. Premiere Pro does not import OMF files."

Basic Method To Export An AAF From Adobe Premiere Pro

It is recommended that the video editor should duplicate the project in Premiere Pro and use that to create the AAF. Then, they should re-organise the project so that only the finalised audio files are included in the sequence.

Let's work our way down the Premiere Pro AAF Export settings window, and we will explain what each one does and our recommended settings.

Mixdown Video

This option will render the video tracks into a single clip that is included

next to the AAF when selected. The video file is an MXF using the DNxHD or DNxHR codec, which should be Pro Tools friendly.

Most people recommend not checking this option and asking the video editor to export an Audio Only AAF from Premiere Pro. However, we have heard reports that this option works with the latest Premiere Pro and Pro Tools versions. But for safety, we still recommend a separate audio-only AAF and a video reference file.

When creating an audio-only AAF, it is recommended that the video editor removes all the video tracks, remembering to unlink the video from the audio first. If they don't, the audio linked to the video tracks will be deleted along with the video.

Finally, they should do a sequence scan for unsupported file characters like /, >, <, ®, and ü, Ø, Æ, Å and rename the files in the AAF Export sequence (not in the bin).

Breakout to Mono

When checked, Premiere Pro will render new audio media rather than link to the existing audio. The AAF and resulting audio media will be in mono with any stereo clips put on mono tracks labelled Left and Right. If this option is left unchecked, Premiere Pro will not render out new audio files and the AAF will link to the existing audio files and maintain the original channel format. Definitely not Pro Tools friendly!

Consequently, this option should *always* be checked.

Render audio clip effects

This option renders any audio effects into the newly created AAF audio files. It is worth checking this option as long as you also check the next option too.

Include clip copies without effects

This option is only available if 'Render audio clip effects' is checked. When checked, Adobe Premiere Pro will include an unprocessed version of any audio file with audio clip effects applied in the Premiere Pro timeline. This means that the AAF will include both a processed and unprocessed version of each file with the copy without effects being placed on a new separate audio track.

Note that both copies will come in the Pro Tools session as unmuted, meaning that both the processed and unprocessed clips will playback together, not desired. As part of your tidying up process, once the AAF has been imported into your Pro Tools session, pick one and mute the other, do not use both. That said, we do recommend that you keep both of them in the session.

In our article, [AAF And OMFs - Post Audio Expert Panel On The Good, The Bad And The Ugly](#), Reid Caulfield says...

"Anecdotally, Premiere Pro AAF's seem to be the most problematic. Missing files, corrupt files, handling of "effected" audio files (i.e. "baked-in" speed ramps, pitch shifts, reverbs etc). We always ask for both the effected ("rendered") file as well as the unrendered files but the unrendered originals are rarely available and the rendered files are corrupt or incompatible with Pro Tools."

Sample Rate

This option sets the sample rate of the newly created audio files. The options are 96K or 48K and we recommend it is set to 48K unless there is a particular reason to use the higher sample rate.

Bits per Sample

This option will set the bit depth of the newly created audio files. We

recommend that this is set to 24 bit unless there are any specific reasons to use a different bit depth.

Files

With the Files option, this is where you decide if your AAF will have the audio files embedded or have separate files that can be linked to.

- Choosing Embed Audio will write the newly created audio files into the AAF export creating a single, self-contained AAF, with the data and audio all wrapped into one file.
- Choosing Separate Audio will cause Premiere Pro to render the newly created audio files into a folder next to the AAF. Subsequent exports using Separate Audio can reuse the exported audio files if they have not changed in the timeline.

The advantage of using Separate Audio means you get more detailed audio metadata. However, it is possible to lose some files or the linking process can go astray.

The advantage of using Embed Audio is that all the audio files are brought together into one AAF file, making it less likely to lose any audio files along the way. However, the metadata gets removed when using the Embed Audio option.

When we asked our Expert Panel in our article [AAF And OMFs - Post Audio Expert Panel On The Good, The Bad And The Ugly](#) Reid Caulfield, Graham Kirkman, Kevin Dallas, Paul Maunder, Tom Lowe and Lucy Mitchell all chose Embed Audio, with comments like these...

"Embedded seems to be working best at the moment, but it depends on the particular version of the NLE in question that they're using"

"Embedded. Explaining to people there are a load of other files they

have to send can get complicated!"

"Unless the project is massive, I prefer embedded audio because it reduces the likelihood that any audio files will be missing or lost. Also, I've seen an issue where non-embedded audio can occasionally have the wrong unique ID, which makes re-linking more difficult when opening the AAF or OMF with Pro Tools. In the worst-case scenario, I've had to re-link using Find All Candidates and re-link every single audio file individually!"

Whereas Korey Pereira prefers Separate Audio saying...

"I have a strong preference for not-embedded AAFs/OMFs. Especially when the editor follows my spec sheet and exports all the audio files as 48/24 .WAV files, it saves the step of needing to "force to session format" on import creating another set of files to keep track of"

The takeaway here is if metadata preservation is important for later relinking in Pro Tools, for example, to be able to use the Field Recorder Workflow in Pro Tools then Separate Audio is the way to go.

Preserve media directory name

This option is only available when Separate Audio is selected. choosing this option will ensure that the newly created media files are placed into a folder structure similar to the folder structure of the original media.

Format

This option enables you to select the audio file format of the newly created audio files. The choices are AIFF or Broadcast Wave. You should always select Broadcast Wave (BWF) especially as this option supports iXML audio metadata.

Render

This option determines how much of the original audio files will be included in the AAF...

- Choose Copy Complete Audio Files to make sure the resulting AAF includes complete, untrimmed audio files in the AAF.
- If you choose Trim Audio Files, Premiere Pro will write new trimmed audio files that are only as long as the portions actually used in the sequence. You can use The Handle Frames (see below) to add a certain amount of audio beyond the edits in the Premiere Pro timeline.

Korey Pereira prefers the Complete Audio Files option vs trimming the audio to a certain handle length...

"Yes, it makes for a larger file size, but makes it MUCH easier to simply drag out the full handles on projects where I don't necessarily have time to do a proper dialogue assembly."

Handle Frames

This option is only available if Trim Audio Files is selected above. Choose this option to render additional audio at the beginning and end of the clips, measured in frames. We recommend a minimum of 6 seconds. Note that the number of frames will depend on the project's frame rate. For example, at 25 fps, 6 seconds will be 25x6 = 150 frames.

Basic Method To Export An AAF From Adobe Premiere Pro

In our article, [AAF And OMFs - Post Audio Expert Panel On The Good, The Bad And The Ugly](#), Soularity Sound's Korey Pereira said this...

"There was a time a few years back when Adobe did something with the way Premiere Pro exports AAF and they started going terribly awry."

AAFs would be completely out of sync and randomly loop audio files. It was bad. For a good 2+ years, I started asking for OMFs from anyone editing in Premiere Pro. This year an editor sent me an AAF from Premiere Pro, which worked fine, but I still request OMFs from Premiere Pro."

So let's work our way down the OMF Export window in Premiere Pro, as we did above with the AAF Export window...

OMF Title

This is where you can name the OMF file that will be created. Note that with AAFs, the AAF takes its name from the Sequence Name.

Sample Rate

This option sets the sample rate of the newly created audio files. The options are 96K or 48K and we recommend it is set to 48K unless there is a particular reason to use the higher sample rate.

Bits per Sample

This option will set the bit depth of the newly created audio files. We recommend that this is set to 24 bit unless there are any specific reasons to use a different bit depth.

Files

From the Files menu, you can choose one of the following...

- **Encapsulate** - With this setting, Premiere Pro exports an OMF file containing the project metadata and all the audio for the selected sequence. Encapsulated OMF files typically are large.
- **Separate Audio** - With this setting, Premiere Pro exports individual mono files into an _omfiMediaFiles folder. The folder name contains the OMF filename.

Render

From the Render menu, you can choose one of the following...

- **Copy Complete Audio Files** - With this setting, Premiere Pro exports complete, untrimmed audio files for each clip used in the sequence, no matter how many times the clip is used nor how many parts of the clip are used.
- **Trim Audio Files** - With this setting, Premiere Pro will write new trimmed audio files that are only as long as the portions actually used in the sequence. You can use The Handle Frames (see below) to add a certain amount of audio beyond the edits in the Premiere Pro timeline.

Handles

In the Handle Frames field, you can specify the length of the handles, in video frames. This amount of time is added to the start and end of the

exported files when you choose Consolidate Media. The default setting is one second, in frames, at the sequence frame rate. If the length of the handles you specify exceeds the length of the clip instance, Premiere Pro exports the whole clip instance.

Include Pan

Our advice is that you leave this unchecked so we do not get any panning information coming across to Pro Tools.

That covers the basic methods for AAF and OMF exports from Adobe Premiere. But what about some of the other *issues* that can come with AAFs coming from Premiere Pro?

Video Tracks Should Not Be Included In the Sequence Used To Create The AAF

The general consensus is that for the AAF export, the video editor should duplicate the sequence and then remove all the video tracks from the duplicate sequence. For example, even with Pro Tools Ultimate 2021.3 and Premiere Pro 2021, we have seen a report where there were issues, not just on import but also when selecting an AAF file in Import Session Data where Pro Tools wouldn't even open the Import Session Data options window. It just threw up a progress bar that said Opening OMF/AAF and Pro Tools beachballed and froze, and Force Quitting Pro Tools was the only way out.

They eventually narrowed it down to the presence of video tracks in the sequence. When the editor removed all the video tracks from his timeline before doing the AAF export, the problem was resolved.

The video editor should do a separate export to create a video file with their guide audio mix included. See below...

Nested Sequences

Check out what Korey Pereira said in our article [AAF And OMFs - Post Audio Expert Panel On The Good, The Bad And The Ugly...](#)

"Nested sequences in Premiere! These are the WORST. There is a certain sub-sect of editors that like to edit their films by creating a sequence for each scene of a film then dropping THAT sequence into a master sequence to assemble (and often further edit) their film.

What this does when you export an AAF, is create a stereo mixdown of each original sequence and export THOSE in the OMF/AAF, which strips the ability to edit any of the audio or even do a proper dialogue assembly as the EDL is just "per scene."

This typically leads to the tough conversation that the editor (or assistant editor) needs to go back and re-assemble the full show or film. For any recurring clients or projects I sign on during pre-production, I have the conversation with editors on the importance of NOT using nested sequences."

In this video, Christopher Campbell from Henninger Media covered what editors need to do when preparing a Premiere Pro project for an AAF export, including making a duplicate sequence and unnesting nested clips...



Merge Clips And Multicam Mode In Adobe Premiere Pro

Premiere Pro provides a method for synchronizing audio and video called Merge Clips. This function streamlines the process by which users can sync audio and video, which have been recorded separately. You can select a video clip and sync it with up to 16 channels of audio by using the Merge Clips command. Clips, which make up the merged clip are referred to as component clips in Premiere Pro.

The consensus is that just like nested clips are a problem, merging clips in Premiere Pro must be avoided if the project is going to be exported as an AAF for separate audio post-production. The forums are full of references to Merged Clips being an issue...

"Merged clips were indeed a big problem in Premiere in regards to metadata coming over."

"Merged is a no-no. Renaming is a no-no. If you can keep the editor inside these parameters you've got a good chance of having an easyish conform."

Adobe confirms this too in their [Synchronizing audio and video with Merge Clips](#) help guide, saying...

"Merged clip audio results in mono track audio-only. Final Cut Pro XML and AAF interchange formats are not supported."

One solution appears to be to use Multicam mode instead, to sync all the audio and video. One user shared this...

"The thing to be careful with is how you treat your audio in your edit, DON'T MERGE CLIPS, we synced the audio and video by creating multicam clips, then at the end of the edit we had to "flatten" all the audio (and video) in the sequences before exporting the relevant files - this is done by highlighting all the multicam clips in the sequence, right clicking and going to "multi-camera" and clicking on "flatten". This changes the synced audio from a synced clips to its original audio source and therefore is the correct file to embed in the AAF."

Another suggestion with regard to Merge Clips is to remove all the video tracks from the sequence before undertaking the AAF Export.

What About The Video?

As we have said above, the advice is to remove all the video tracks from the duplicate Premiere Pro sequence, being used to create the AAF or OMF.

Consequently, as well as the AAF or OMF, you will need a separate reference video file. When exporting the video file make sure the following options are selected...

- It has a burnt in timecode (BITC) with a transparent or translucent window behind it, usually, a few lines up from the middle of the bottom of the picture works best.

- It is encoded with either the Avid DNxHD or ProRes Proxy codec. Make sure you or the video editor avoid codecs such as H264 as they are notoriously unreliable in older versions of Pro Tools.
- Set the resolution to 1920×1080 unless there's a specific file size constraint in your system. In that case, go for 720p (1280×720). You can opt for 4k if your system supports it but we don't recommend it.
- Include the audio mix from Premiere Pro in the reference video as 48 kHz, 24 bit .wav. This helps to provide an idea of the director's intentions for the mix and enable us to check sync against the AAF.

In Conclusion

Overall, it seems that more recently, AAF has become the method of choice, with the key decision being whether to use Embed Audio or Copy Complete Audio Files. It seems that Copy Complete Audio Files is better for metadata flow, but comes with the risk of not getting everything, whereas Embed Audio is a safer bet as long as you don't need the metadata to use with the Field Recorder workflow in Pro Tools.

If your experience differs, please share them in the comment below, but please do state what version of Premiere Pro and Pro Tools are being used because things have changed over the years.

Mike Thornton has been involved in the broadcast audio industry for all his working life, some 45 years. Mike has worked with Pro Tools since the mid-1990s recording, editing and mixing documentaries, comedy and drama for both radio and TV as well as doing the occasional music project. He was the co-founder of Pro Tools Expert and has now retired and has taken up the role of Chairman of Production Expert Ltd.

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